AI Waste Classification & Recycling Assistant

Mini Project Documentation

Project Type	AI-assisted waste awareness and recycling web application
AI Development Tool	Base44 AI app builder
Live Project	petite-eco-vision-sort.base44.app
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Document purpose	Academic mini-project report and GitHub repository documentation

A practical concept for helping users identify waste categories and make more informed recycling decisions.

Abstract

AI Waste Classification & Recycling Assistant is a web-based mini-project intended to support better everyday waste sorting. It focuses on the common challenge of deciding whether an item should be recycled, composted, or handled as general waste. The project was created with Base44, an AI app-building platform that can generate application screens and workflows from natural-language instructions. The published application is available at the project link on the cover.

This document describes the project idea, objectives, proposed workflow, system design, expected benefits, testing plan, limitations, and future improvements. Because the live app's internal implementation and feature behavior were not independently inspectable during preparation, feature-specific statements are framed as intended functionality and should be confirmed against the deployed app before submission.

1. Introduction

Waste segregation is an important part of reducing landfill waste and improving recycling outcomes. However, people may be uncertain about how to dispose of packaging, food scraps, paper, glass, metals, batteries, and other household items. Incorrect sorting can contaminate recycling streams or cause unsafe disposal.

The proposed assistant provides a digital, user-friendly way to learn about waste categories and disposal practices. It can serve as an educational companion for students, households, and communities interested in sustainable habits.

2. Problem Statement

People often lack quick, clear guidance for sorting everyday waste. Disposal rules also vary by location and by local waste collection systems. A simple digital assistant can help explain likely categories and recommend safer next steps, while reminding users to verify local rules for special or hazardous waste.

3. Objectives

• Create an accessible web application focused on waste classification and recycling awareness.

• Help users understand common waste categories and suitable disposal actions.

• Promote responsible habits such as separating recyclable, organic, residual, and special waste.

• Use an AI app-building tool to accelerate interface creation and project iteration.

• Provide a foundation that can later be expanded with image recognition, local recycling guidance, and usage analytics.

4. Scope

The project is designed as an educational support tool, not a replacement for municipal waste instructions or professional hazardous-waste advice. Its usefulness depends on the features implemented in the live application and the accuracy of any classification or recommendations it provides.

Intended user groups

Students, households, teachers, community groups, and people who want quick guidance on common waste items.

Typical waste categories

Paper and cardboard; plastics; glass; metals; organic or compostable waste; general/residual waste; and special items such as batteries, e-waste, chemicals, or medical waste that may require designated collection.

5. Tools and Technologies

Tool / component	Role in the project
Base44 AI app builder	AI-assisted creation and iteration of the web application; platform capabilities can include screens, workflows, data, and deployment.
Web browser	Accessing and testing the published application.
GitHub	Version control, source-code storage (if exported), issue tracking, and project documentation.

Note: The exact framework, database, model, and integrations used by this particular deployment were not confirmed. Add them only after checking the Base44 project editor or exported source code.

6. Proposed System Workflow

• User opens the published web application.

• User selects, enters, or provides information about a waste item, depending on the interface available.

• The application determines or presents a likely waste category using its implemented logic.

• The application displays guidance about recycling, composting, or disposal, if supported.

• User checks local collection rules, especially for batteries, electronics, chemicals, medical waste, and contaminated packaging.

The sequence above is a conceptual workflow. Confirm the exact inputs and outputs in the deployed application before describing them as implemented features.

7. High-Level Architecture

User	Web interface	Application logic	Guidance / result
Enters or selects an item	Form or page presented by the app	Configured Base44 app logic and any connected services	Category and disposal advice, where implemented

Base44 is the development platform. Do not assume that an image-classification machine-learning model is connected unless this is verified in the app configuration or source code.

8. Implementation Approach

8.1 Define the concept

Describe the waste-sorting problem, target users, and expected user journey.

8.2 Generate the application

Use natural-language prompts in Base44 to create an initial interface and core workflow.

8.3 Refine the experience

Review wording, page layout, accessibility, mobile behavior, and guidance clarity.

8.4 Validate the published app

Test representative waste examples, error states, and navigation using the live URL.

8.5 Document and share

Record the project overview, setup or access instructions, limitations, and screenshots in GitHub.

9. Testing Plan

Test	Expected outcome
Open live URL	Application loads without a blocking error.
Common recyclable item	The app gives a relevant category or useful guidance, if supported.
Organic waste item	The app handles the example consistently with its intended rules.
Ambiguous or mixed-material item	The app communicates uncertainty or recommends checking local rules.
Hazardous/special item	The app avoids unsafe general advice and points users to designated collection guidance.
Mobile viewport	Text and controls remain readable and usable.
Invalid or empty input	The interface handles missing input clearly, if user input is supported.

Record actual results (Pass/Fail and notes) after testing. Do not report tests as passed until they have been run.

10. Benefits

• Encourages awareness of waste segregation and recycling.

• Makes basic disposal guidance easier to access through a web link.

• Demonstrates how AI-assisted app-building can support rapid prototyping.

• Can be extended for local rules, multiple languages, and community education.

11. Limitations and Safety Considerations

• Waste rules differ between cities, facilities, and collection providers.

• Items made from multiple materials may require special handling or separation.

• AI-generated suggestions may be incomplete or incorrect; users should verify important disposal decisions.

• Batteries, e-waste, chemicals, medicines, sharps, and medical waste should not be treated as ordinary recyclables.

• Image recognition accuracy, if available, must be tested on different lighting, angles, damaged items, and visually similar materials.

• The project should not claim environmental impact reductions without measured evidence.

12. Future Enhancements

• Add or improve image-based waste recognition and show confidence or uncertainty.

• Provide location-aware recycling instructions sourced from trusted local authorities.

• Add multilingual guidance and accessible voice input/output.

• Include clear instructions for e-waste, batteries, and other special waste streams.

• Add a feedback mechanism for users to flag incorrect recommendations.

• Track anonymized usage and evaluation results only with appropriate privacy safeguards.

13. Conclusion

AI Waste Classification & Recycling Assistant is a sustainability-focused mini-project that explores how an AI-assisted web application can make waste-sorting information easier to access. Base44 helped provide a route from a natural-language concept to a published application. The next important step is to validate the live app's actual features and classification quality, then document measured test results and local disposal guidance.

14. Project Links

Live application: https://petite-eco-vision-sort.base44.app

Base44 AI app builder: https://base44.com/ai-app-builder

Base44 CLI / developer resources: https://github.com/base44/cli

The live project link is included as supplied. The implementation details in this report are intentionally general where the deployed application's internal behavior could not be confirmed.

Appendix A — Suggested GitHub README

Use the following text in the repository's README.md, then update the feature list and testing status to match the actual app.

````
```markdown
# AI Waste Classification & Recycling Assistant

A mini project built with the help of Base44 AI to promote waste segregation and
recycling awareness.

## Live Demo
https://petite-eco-vision-sort.base44.app

## Overview
The project aims to help users understand common waste categories and make more
informed disposal decisions. Always follow local waste-management guidance, especially
for hazardous and electronic waste.

## Built With
- Base44 AI app builder
- Web browser for access and testing

## Project Documentation
See `AI_Waste_Classification_Recycling_Assistant_Documentation.pdf` for the project
report.

## Testing
Add the test cases performed and their actual results here. Do not mark tests as passed
before running them.

## Limitations
Recommendations may vary by location and item composition. Verify local disposal rules.

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```
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Appendix B — Suggested GitHub Publishing Steps

• Create a new GitHub repository named `AI-Waste-Classification-Recycling-Assistant`.

• Add this PDF and a README.md. Add screenshots of the actual application if permitted.

• If you want to publish source code, export or sync it from Base44 using the options available to your account, then review the files before committing.

• Never commit API keys, passwords, private user data, or other secrets. Add appropriate files to `.gitignore`.

• In the repository description, include the purpose of the project and the live demo URL.

• Verify that the PDF opens correctly and that the live link works from the repository page.

Reference: Base44 describes its platform as an AI app builder for creating and publishing web applications. GitHub/source-code export options may depend on the account plan and current platform features.